

Industrial Internship Report on Spotify Clone – Web Based Music Player

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Executive Summary

This report presents the details of the Industrial Internship provided by **upskill Campus** and **The IoT Academy** in collaboration with the industrial partner **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship duration was **6 weeks**, during which we were assigned an industry-oriented project and asked to complete both development and documentation within the given timeline.

My assigned project was **“Spotify Clone – Web Based Music Player”**, developed using **HTML, CSS, and JavaScript**. The project focuses on building a user-friendly music player that allows users to play, pause, seek, and navigate songs through an interactive interface.

This internship gave me practical exposure to real-world development practices, improved my problem-solving skills, and enhanced my understanding of frontend technologies. Overall, it was a valuable and enriching learning experience.

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1. Preface

This internship was a **6-week industrial training program** focused on providing hands-on experience in real-world projects. Internships play a very important role in career development as they help students apply theoretical knowledge to practical problems.

My project, **Spotify Clone**, was based on frontend web development. The opportunity provided by **upskill Campus (USC)** and **UniConverge Technologies Pvt Ltd (UCT)** helped me understand industry expectations, project planning, and structured development.

The program was well planned with regular guidance, milestones, and evaluations. During this internship, I improved my technical skills, communication, and confidence. I would like to thank **upskill Campus, The IoT Academy, UCT mentors**, and my peers for their continuous support. I encourage my juniors to actively participate in such internships for better career growth.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT), established in **2013**, works in the **Digital Transformation** domain. The company provides industrial solutions with a strong focus on **sustainability** and **return on investment (RoI)**.

UCT uses modern technologies such as:

- Internet of Things (IoT)
- Cyber Security
- Cloud Computing (AWS, Azure)
- Machine Learning
- Communication Technologies (4G/5G/LoRaWAN)
- Java Full Stack, Python, and Frontend technologies

i. UCT IoT Platform (Insight)

UCT Insight is an IoT platform developed using **Java (Backend)** and **ReactJS (Frontend)**. It supports **MySQL** and **NoSQL** databases.

Features:

- Device connectivity using MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- Cloud and on-premises deployment
- Custom dashboards
- Analytics and reporting
- Alerts and notifications

- Third-party integration (SAP, Power BI, ERP)
- Rule Engine

ii. Smart Factory Platform

Factory Watch is designed for smart factory solutions such as:

- Production and asset monitoring
- OEE and predictive maintenance
- KPI monitoring and improvement
- Modular and scalable SaaS architecture

iii. LoRaWAN Based Solutions

UCT provides solutions for **Smart Cities, Agriculture, Smart Street Lights, and Smart Metering** using LoRaWAN technology.

iv. Predictive Maintenance

UCT offers machine health monitoring and predictive maintenance using **Industrial IoT** and **Machine Learning**.

2.2 About upskill Campus (USC)

upskill Campus, along with The IoT Academy and UniConverge Technologies, ensures smooth execution of the internship program. USC is a **career development platform** that provides affordable and scalable learning solutions with measurable outcomes.

2.3 The IoT Academy

The IoT Academy is the EdTech division of UCT. It conducts executive certification programs in collaboration with **EICT Academy, IIT Kanpur, IIT Roorkee, and IIT Guwahati**.

2.4 Objectives of this Internship Program

The objectives of this internship were:

- To gain practical industry experience
 - To solve real-world problems
 - To improve job opportunities
 - To enhance technical and domain knowledge
 - To develop communication and problem-solving skills
-

2.5 Reference

- [1] upskill Campus Learning Material
- [2] JavaScript Documentation
- [3] MDN Web Docs

2.6 Glossary

Term	Acronym
Hyper Text Markup Language	HTML
Cascading Style Sheets	CSS
JavaScript	JS
User Interface	UI

3. Problem Statement

The problem statement was to design and develop a **web-based music player** that provides core features similar to Spotify. The system should allow users to play, pause, navigate songs, and visually track playback progress using a simple and responsive interface.

4. Existing and Proposed Solution

Existing Solution

Most existing music players are either desktop-based or complex web applications that require backend services and user authentication.

Limitations:

- Complex architecture
- Requires server-side support
- Not suitable for beginners

Proposed Solution

The proposed solution is a **frontend-only Spotify Clone** using HTML, CSS, and JavaScript.

Value Addition:

- Simple and lightweight design

- No backend dependency
- Easy to understand and modify
- Responsive UI

4.1 Code Submission (GitHub Link)

<https://github.com/Intiyaj-Raj/upskillcampus>

4.2 Report Submission (GitHub Link)

(Placeholder – same repository)

5. Proposed Design / Model

The project follows a simple flow:

- User selects a song
- JavaScript audio object loads the file
- Play/Pause and navigation controls manage playback
- Progress bar updates in real time

5.1 High Level Diagram

Figure 1: High Level Architecture of Spotify Clone
(User Interface → JavaScript Logic → Audio Files)

5.2 Low Level Diagram

UI Controls → Event Listeners → Audio API → Output Sound

5.3 Interfaces

- User Interface: HTML & CSS
- Logic Interface: JavaScript
- Media Interface: Audio files (MP3)

6. Performance Test

Constraints

- Browser compatibility
- Audio loading time
- UI responsiveness

Handling Constraints

- Optimized images
- Lightweight JavaScript
- Tested on multiple browsers

6.1 Test Plan / Test Cases

- Play/Pause button test
- Song navigation test
- Progress bar seek test

6.2 Test Procedure

Manual testing using Chrome and Firefox browsers.

6.3 Performance Outcome

The application performed smoothly with minimal delay and good responsiveness.

7. My Learnings

- Strong understanding of HTML, CSS, and JavaScript
- Audio handling using JavaScript
- UI/UX design basics
- Project structuring and documentation

This internship will help me significantly in my **frontend development career**.

8. Future Work Scope

- Add backend support
- User authentication
- Online streaming support
- Playlist creation feature
- Mobile app version